GEO1016 Assignment 2: Triangulation

1 Introduction

In this assignment, you will gain hands-on experience in reconstructing 3D geometry from the corresponding image points of two images. Specifically, you will practice:

- Basic linear algebra operations.
- Estimating fundamental matrix F from corresponding image points.
- Recovering relative pose (i.e., R and $\mathbf{t}$) from the fundamental/essential matrix.
- Determining 3D coordinates of image points using the recovered relative pose.

2 Methodology

The underlying theory has been explained in the lecture and is elaborated in the lecture notes (i.e., handout). In this section, mainly the major steps of each algorithm are provided. Please refer to the lecture notes and/or the lecture slides for the technical details.

Step 1: Estimate fundamental matrix F from ALL provided corresponding image points using the normalized 8-point algorithm (20%)

- Normalization. (5%)
- Linear solution (based on SVD). (5%)
- Constraint enforcement (based on SVD). (5%)
  - Find the closest rank-2 matrix.
- Denormalization. (5%)

Step 2: Recover relative pose (i.e., R and $\mathbf{t}$) from the fundamental matrix (20%)

- Find the 4 candidate relative poses (based on SVD). (10%)
- Determine the correct relative pose. (10%)
  - The correct $(R, \mathbf{t})$ pair is the one that yields the most (in theory, it is 'all' but may not in practice due to noise) 3D points lying in front of both cameras (i.e., z values w.r.t. both cameras are positive). See Step 3 for 3D geometry recovery.

Step 3: Reconstruct 3D coordinates for all corresponding image points (20%)

- Compute the projection matrix from K, R, and $\mathbf{t}$. (5%)
- Triangulate all corresponding image point pairs using the linear method. (15%)
- [Optional] Non-linear least-squares refinement of the 3D points computed from the linear method. You must refine the coordinates of all points simultaneously (instead of one by one by calling the same function multiple times). A example of solving non-linear least squares problem using the Levenberg-Marquardt method is given in `/A2_Triangulation_Code/Example_NonlinearLeastSquares`. (10% extra)

Step 4: Evaluation (10%)

- Come up with a method to evaluate the quality of the reconstructed 3D points (5%).
- Invite student(s) of another group to review your code. Incorporate the received feedback in your implementation and reflect on the changes in the report. (5%).

It is strongly advised to define a function for each sub-task. This way you have a clean and well-structured implementation, which also makes your testing and debugging easier. Each function may also be reused for other purposes.

Note: You're encouraged to discuss and share experiences with other groups (but copying code from others will be penalized) and discuss what you have learned in the report.

3 Data, Code Framework, and Viewer

The data used in this assignment are 160 pairs of image points from two images. The points from the two images have one-to-one correspondences. All points are stored in homogenous coordinates, i.e., x, y, w (and w is set to 1.0) in each line representing a single point. The points of the two images are stored separately in:

- /A2_Triangulation_Code/resources/data/image_points_0.xyz
- /A2_Triangulation_Code/resources/data/image_points_1.xyz

Build and run the viewer from the provided source code, the image points will be automatically loaded for you (see Figure 1). If you press the `space` key, the reconstruction method will be triggered. The reconstruction function is named `triangulation(...)` and is located in `A2_Triangulation_Code/Triangulation/triangulation_method.cpp`. You will need to implement this function to reconstruct 3D points from the corresponding image points. Comments and guidance are provided in this function. Example code is also provided within this function for you to get familiar with the necessary data structures and APIs.

In addition to the interactions (e.g., rotate, translate, zoom in/out) provided by the viewer, you can also

- take a screenshot of the visualization by pressing the 's' key. (It is required to include in your report a few screenshots of the reconstructed 3D points from different views).
- save the reconstructed 3D points into a file by pressing 'Ctrl' and 's'. (It is required to include the file of your reconstructed 3D points in your submission).

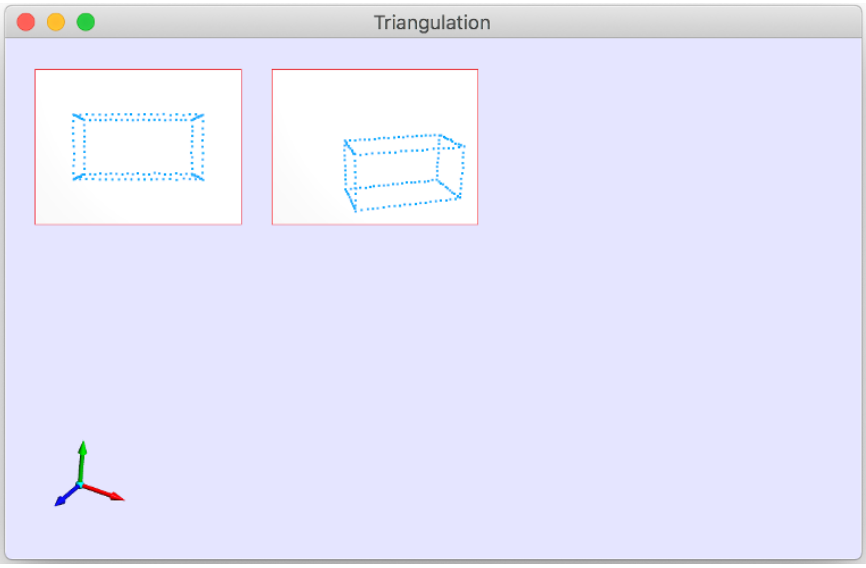
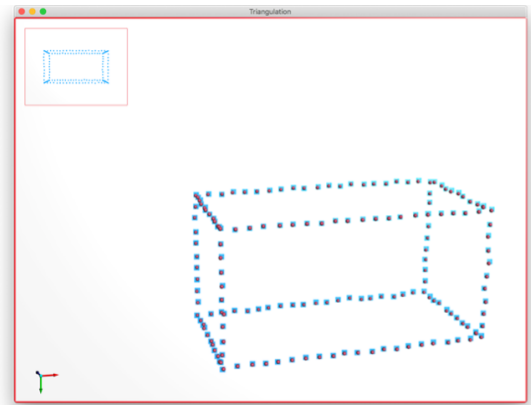
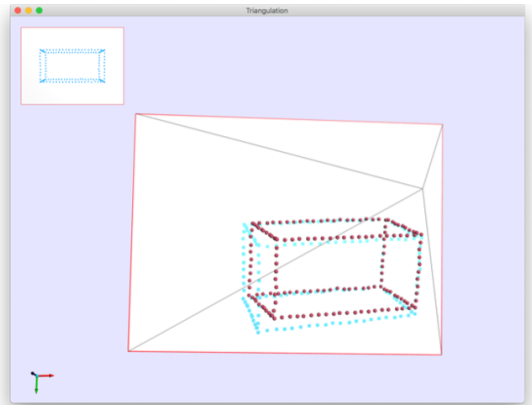


Figure 1: The two images and the input image points.



(a) 3D points visualized using R and $\mathbf{t}$



(b) 3D points visualized in another view

Figure 2: Your reconstruction result should look like this.

When the `triangulation(...)` function has been correctly implemented, running the method will also automatically update the viewer so the reconstructed points will be visualized using the recovered camera extrinsic parameters. Such visualization can help you validate your result (see Figure 2).

4 Submission

Your submission should include:

1) **Report (30%)** (Max 3 pages excluding figures and tables).

   You're expected to deliver a clear, precise, and well-structured scientific report. Make every sentence, equation, and notation precise and clear to avoid misinterpretation. Meanwhile, the report should be as concise as possible, but it should provide key information to reimplement the method to reproduce your results, and it should include:

   - Description of the methodology. We encourage the use of mathematical languages (e.g., equations). (5%)
   - Demonstration of your result. (5%)

- Discussion on the quality of the reconstruction. Is there a way to quantitatively measure its accuracy? (5%)
- Possible improvements. Can you still improve the reconstruction result (assuming the camera intrinsics and image matching are both accurate)? (5%)
- Effects of inaccuracies in camera parameters. In this assignment, the ground-truth intrinsic parameters are given. In practice, how can you obtain accurate camera intrinsic parameters? Introduce a certain level of inaccuracy (e.g., 5%, 10%, 20%) in these parameters (in the function `Triangulation::key_press_event(...)` in `triangulation.cpp`) and see if you can still obtain a reasonably good reconstruction result. What can you conclude from these tests? (5%)
- Reflection on feedback from the other group. How did it help or improve your implementation? (5%)
- Workload distribution (i.e., who did what) using a few bullet points for each member.

To promote formal scientific writing, the following rules apply for assessment:

- Any misunderstanding or misconception of main concepts: 10% deduction.
- Multiple unclear or ambiguous descriptions: 10% deduction.
- Multiple typos, grammar issues, format issues (e.g., a figure/table without a caption or multiple figures/tables with the same caption, unindexed or unreferenced figures/tables), consistency issues (e.g., upper case and lower case used interchangeably, regular font and italic font used interchangeably), misuse of symbols in notations: 10% deduction.
- If the report exceeds the length limit by more than half a page: 10% deduction.

2) **Data**. An `.xyz` file containing the reconstructed 3D points. You can save the reconstructed 3D points into such a file by pressing 'Ctrl' and 's'.

3) **Source code**. Only the following file is required:

   - `A2_Triangulation_Code/Triangulation/triangulation_method.cpp`

   Note: your source code should compile and reproduce your results without modification. Additionally, please try to test your code on different operating systems (if available), as it may be evaluated on a system different from the one used for your development.

How to submit? Compress all required files into a single archive named in the following format:

GEO1016_A2_Group_XX.zip

where "XX" is the 2-digit ID of your group ID (e.g., 03). If you submit an updated version, append a version number, e.g., "_v2", to the file name.